

Proof. 1. **Vector notation.** Define

$$\mathbf{a} = (a_1, \dots, a_n), \quad \mathbf{b} = (b_1, \dots, b_n) \in \mathbb{R}^n.$$

Their Euclidean inner product and norm are

$$\mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^n a_i b_i, \quad \|\mathbf{a}\|^2 = \sum_{i=1}^n a_i^2, \quad \|\mathbf{b}\|^2 = \sum_{i=1}^n b_i^2.$$

Thus we have to show

$$(\mathbf{a} \cdot \mathbf{b})^2 \leq \|\mathbf{a}\|^2 \|\mathbf{b}\|^2. \quad (1)$$

2. **Degenerate case.** If $\|\mathbf{b}\|^2 = 0$ then $b_i = 0$ for every i , so $\mathbf{a} \cdot \mathbf{b} = 0$ and both sides of (1) are zero. By symmetry the same holds when $\|\mathbf{a}\|^2 = 0$. Henceforth we assume

$$\|\mathbf{a}\|^2 > 0, \quad \|\mathbf{b}\|^2 > 0. \quad (2)$$

3. **Quadratic auxiliary function.** For $t \in \mathbb{R}$ set

$$f(t) = \sum_{i=1}^n (a_i - t b_i)^2. \quad (3)$$

Each summand is a square, therefore $f(t) \geq 0$ for all $t \in \mathbb{R}$.

Expanding a typical term,

$$(a_i - t b_i)^2 = a_i^2 - 2t a_i b_i + t^2 b_i^2,$$

we obtain

$$f(t) = \sum_{i=1}^n a_i^2 - 2t \sum_{i=1}^n a_i b_i + t^2 \sum_{i=1}^n b_i^2.$$

Thus

$$f(t) = At^2 + Bt + C, \quad (4)$$

with

$$A = \sum_{i=1}^n b_i^2 > 0, \quad B = -2 \sum_{i=1}^n a_i b_i, \quad C = \sum_{i=1}^n a_i^2. \quad (5)$$

4. **Discriminant condition.** A quadratic $At^2 + Bt + C$ with $A > 0$ is non-negative for every real t iff its discriminant does not exceed zero:

$$B^2 - 4AC \leq 0. \quad (6)$$

Applying (6) to (4)–(5) yields the desired inequality.

5. **Compute the discriminant.** Substituting (??) into (??) gives

$$\begin{aligned} B^2 - 4AC &= \left(-2 \sum_{i=1}^n a_i b_i\right)^2 - 4 \left(\sum_{i=1}^n b_i^2\right) \left(\sum_{i=1}^n a_i^2\right) \\ &= 4 \left[\left(\sum_{i=1}^n a_i b_i\right)^2 - \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right) \right]. \end{aligned}$$

Dividing the inequality $B^2 - 4AC \leq 0$ by the positive constant 4 we obtain exactly (??). Hence the Cauchy–Schwarz inequality holds.

6. **Equality condition.** Equality in (??) occurs precisely when the discriminant in (??) equals zero, i.e.

$$B^2 - 4AC = 0. \quad (7)$$

For a quadratic with $A > 0$, discriminant zero means that there exists $t_0 \in \mathbb{R}$ such that $f(t_0) = 0$. Because $f(t)$ is a sum of non-negative squares (??), $f(t_0) = 0$ forces each term to vanish:

$$a_i - t_0 b_i = 0 \quad \text{for every } i,$$

hence

$$a_i = t_0 b_i \quad (i = 1, \dots, n). \quad (8)$$

Thus $\mathbf{a} = \lambda \mathbf{b}$ with $\lambda = t_0$.

Conversely, if $a_i = \lambda b_i$ for some $\lambda \in \mathbb{R}$, then

$$\sum a_i b_i = \lambda \sum b_i^2, \quad \sum a_i^2 = \lambda^2 \sum b_i^2,$$

and a direct substitution shows

$$\left(\sum a_i b_i\right)^2 = \lambda^2 \left(\sum b_i^2\right)^2 = \left(\sum a_i^2\right) \left(\sum b_i^2\right).$$

The degenerate case $\lambda = 0$ corresponds to both vectors being zero. Therefore equality in the Cauchy–Schwarz inequality holds *iff* there exists $\lambda \in \mathbb{R}$ such that $a_i = \lambda b_i$ for all i .

□